

Hotel Booking Cancellation Analysis

An End-to-End Data Analysis Project

Dataset: Hotel Bookings (City Hotel & Resort Hotel)

Tools used: Python (Pandas, NumPy, Matplotlib, Seaborn)

1. Business Problem

In recent years, City Hotel and Resort Hotel have seen high cancellation rates. Each hotel is now dealing with a number of issues as a result, including fewer revenues and less than ideal hotel room utilisation. Consequently, lowering cancellation rates is both hotels' primary goal in order to increase their efficiency in generating revenue, and for us to offer thorough business advice to address this problem.

The analysis of hotel booking cancellations, as well as other factors that have a bearing on the hotels' business and yearly revenue generation, are the main topics of this report.

2. Hypothesis

1. More cancellations occur when prices (Average Daily Rate) are higher.
2. When there is a longer waiting list, customers tend to cancel more frequently.
3. The majority of clients are coming through Online Travel Agents (OTA) to make their reservations.
4. The biggest factor affecting the effectiveness of earning income is booking cancellations.
5. Cancellations result in vacant rooms for the booked length of time.
6. Clients make hotel reservations in the same year they make cancellations.

3. Assumptions

1. No unusual occurrences between 2015 and 2017 will have a substantial impact on the data used.
2. The information is still current and can be used to analyze a hotel's possible plans in an efficient manner.
3. There are no unanticipated negatives to the hotel employing any of the suggested techniques.
4. The hotels are not currently using any of the suggested solutions.

4. Research Questions

1. What are the variables that affect hotel reservation cancellations?
2. How can hotel reservation cancellations be reduced?
3. How can hotels be assisted in making better pricing and promotional decisions?

5. Analysis and Findings

5.1 Overall Reservation Status

Out of all the reservations recorded in the dataset, about 62.9% were not cancelled and 37.1% were cancelled. This shows that more than a third of all bookings end up being cancelled, which has a direct impact on hotel revenue and room occupancy planning.

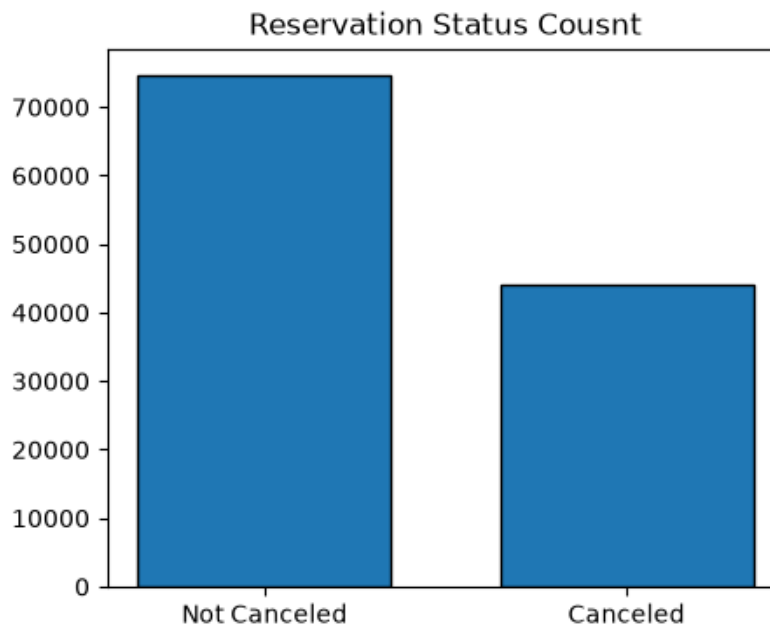


Figure 1: Overall reservation status (cancelled vs. not cancelled)

5.2 Cancellations by Hotel Type

Splitting the data by hotel type shows a clear difference in cancellation behaviour:

- Resort Hotel: about 28% of reservations were cancelled (72% honoured).
- City Hotel: about 41.7% of reservations were cancelled (58.3% honoured).

City Hotel therefore has a noticeably higher cancellation ratio than Resort Hotel, even though it also receives the larger share of total bookings.

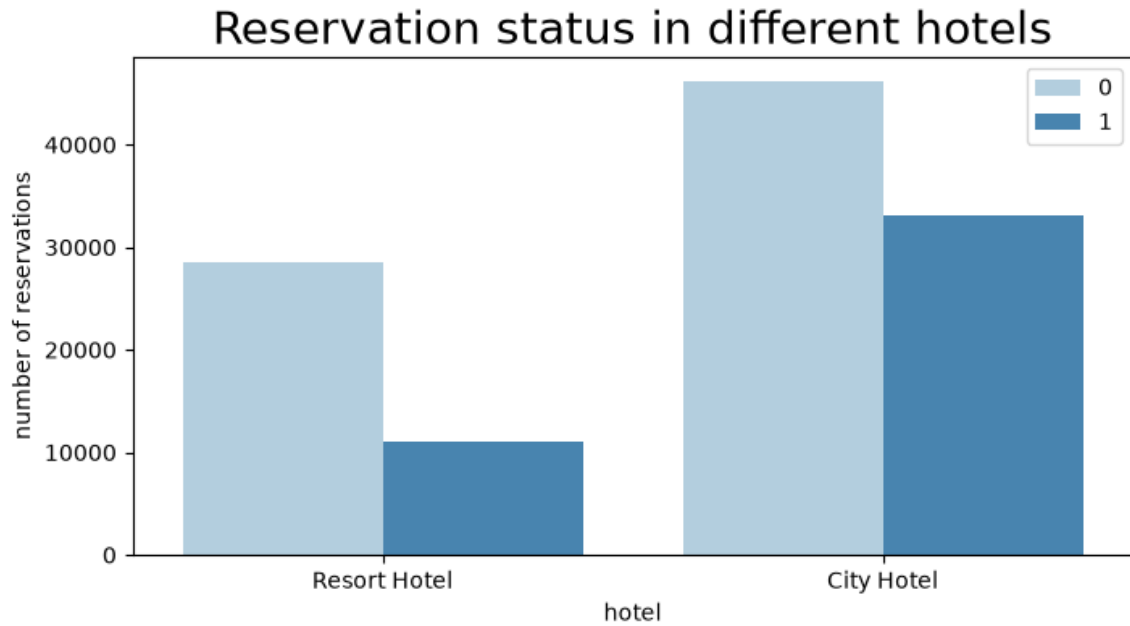


Figure 2: Number of reservations (cancelled vs. not cancelled) by hotel type

5.3 Price (ADR) vs. Cancellations

Comparing the Average Daily Rate (ADR) trend for City Hotel and Resort Hotel over time shows that Resort Hotel rates fluctuate more sharply, with several high seasonal peaks, while City Hotel rates are comparatively steadier. City Hotel also receives more bookings overall, suggesting that Resort Hotel rooms may be priced higher on average, especially during peak periods.

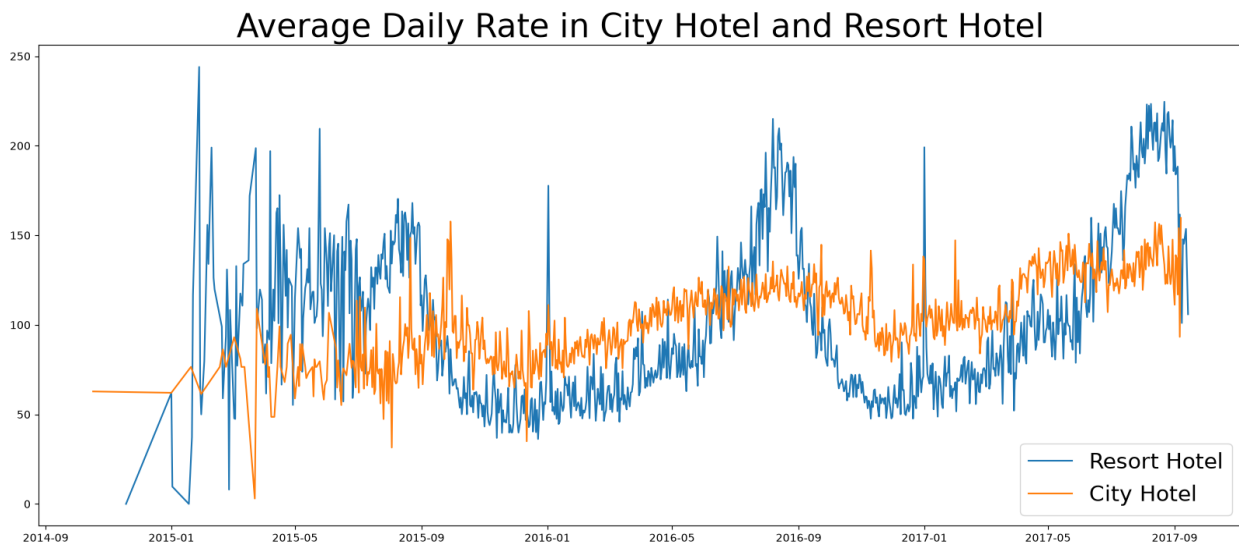


Figure 3: Average Daily Rate (ADR) over time – City Hotel vs. Resort Hotel

When the average daily rate of cancelled bookings is compared with non-cancelled bookings (2016–2017), the cancelled bookings consistently show a higher ADR for most of the period. This supports Hypothesis 1: reservations made at a higher price point are more likely to be cancelled than those made at a lower price point.

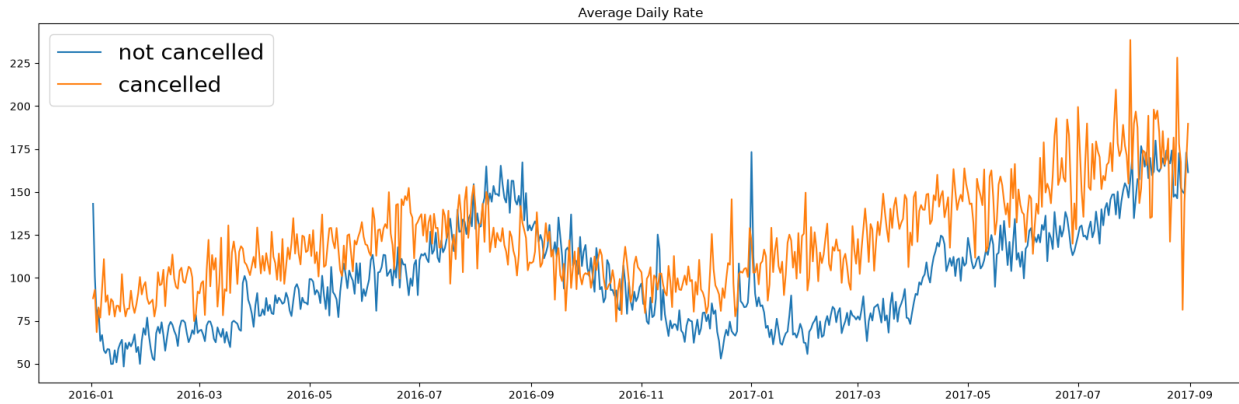


Figure 4: Average Daily Rate – cancelled vs. not cancelled (2016–2017)

5.4 Cancellations by Month

Reservation status was analysed on a month-by-month basis to identify seasonal patterns. January records the highest number of cancellations of any month, while the months from July to October show the highest volume of confirmed (not cancelled) bookings and comparatively low cancellation counts.

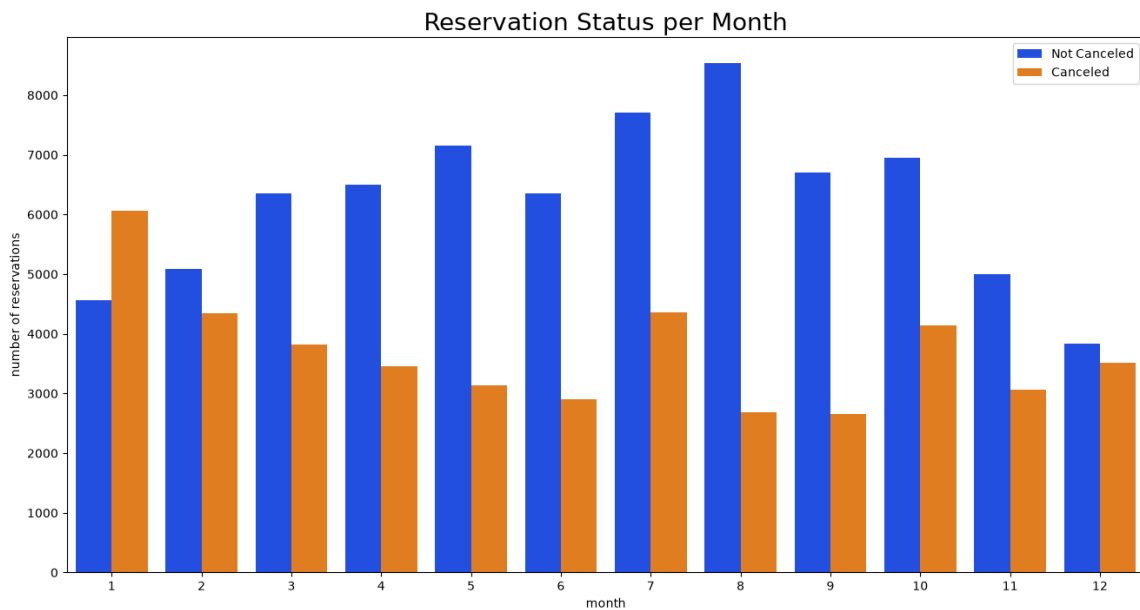


Figure 5: Reservation status by month (cancelled vs. not cancelled)

Looking specifically at the total ADR generated by cancelled bookings each month, January again stands out with the highest combined ADR value among cancelled reservations, reinforcing that January is the most critical month for revenue lost to cancellations.

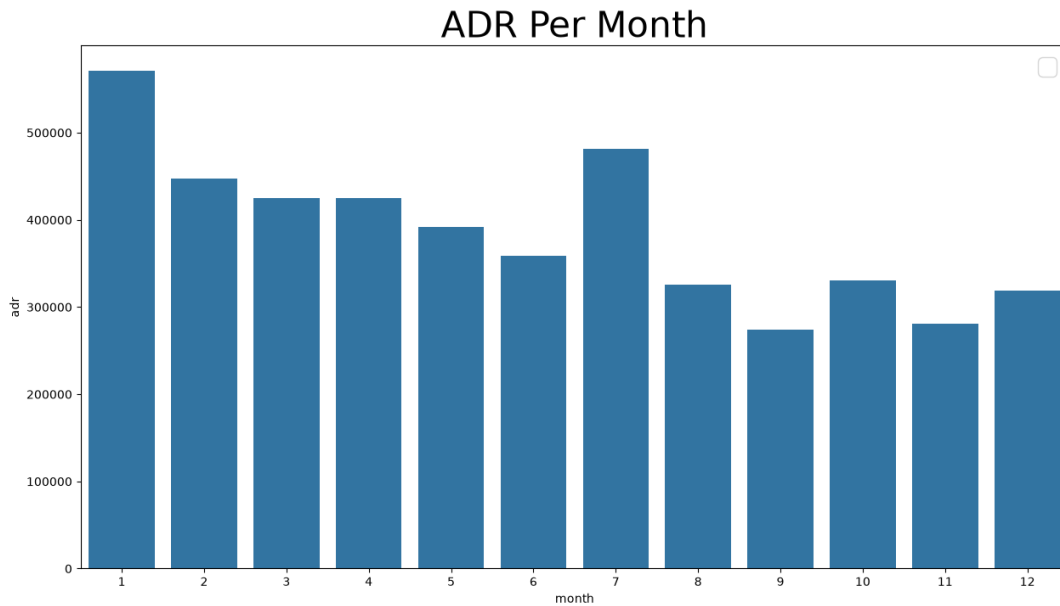


Figure 6: Total ADR of cancelled bookings, by month

5.5 Cancellations by Country

Among guests who cancelled their reservations, Portugal (PRT) accounts for by far the largest share, followed by the United Kingdom (GBR), Spain (ESP), France (FRA), Italy (ITA), Germany (DEU), Ireland (IRL), Brazil (BRA), the USA and Belgium (BEL) rounding out the top 10.

Top 10 Countries with Reservation Cancelled

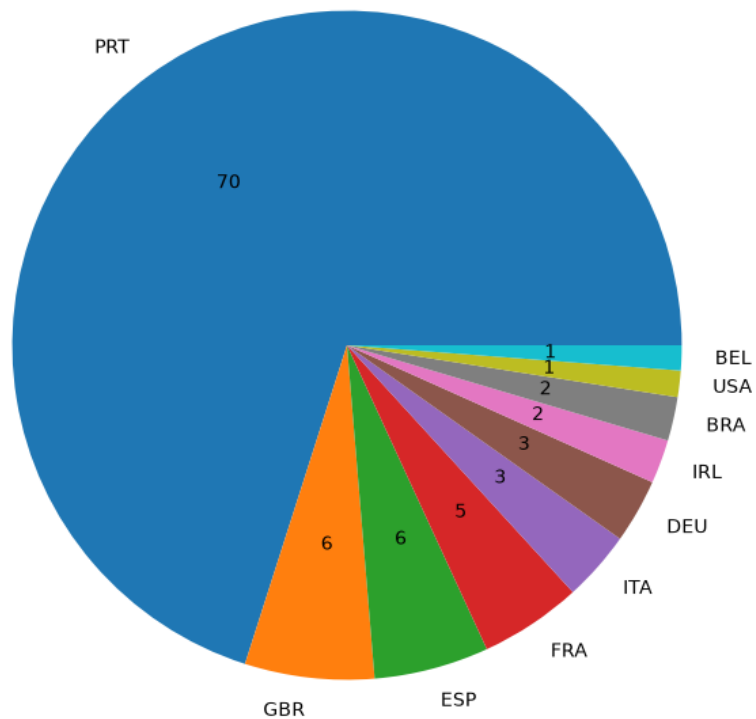


Figure 7: Top 10 countries by number of cancelled reservations

5.6 Booking Source (Market Segment)

Looking at where reservations originate from overall, around 47% of all bookings come through Online Travel Agents (OTA), about 20% through Offline Travel Agents/Tour Operators, roughly 17% through Groups, about 10.5% are Direct bookings, and the remainder come through Corporate, Complementary and Aviation channels.

However, when looking only at the bookings that were cancelled, the mix shifts: Online TA still leads at around 47%, but Groups jump to about 27% of cancellations (compared to 17% of all bookings), while Offline TA/TO drops to about 19% and Direct bookings fall to roughly 4%. This indicates that bookings made through Groups and Online Travel Agents are disproportionately more likely to be cancelled compared with Direct or Offline Travel Agent bookings.

6. Suggestions / Recommendations

1. Cancellation rates rise as the price increases. To reduce cancellations, hotels could review their pricing strategy and consider lowering rates for specific locations, or offer targeted discounts to price-sensitive customers.
2. Since the cancellation ratio is higher for City Hotel than for Resort Hotel, City Hotel should consider offering reasonable discounts on room prices, particularly on weekends or holidays, to reduce cancellations and improve occupancy.

3. Because January has the highest number of cancellations and the highest cancelled-booking ADR, hotels should plan targeted marketing campaigns or promotional offers ahead of January to retain bookings and offset potential revenue loss.
4. Hotels should focus on improving overall service quality, particularly in Portugal (the largest source of cancellations), and strengthen relationships with Group and Online Travel Agent channels, since these segments show the highest cancellation propensity.
5. A flexible or partially refundable booking policy for high-ADR reservations could reduce the incentive for guests to cancel when they find a cheaper alternative closer to the stay date.

7. Conclusion

This analysis confirms that booking cancellations are a significant driver of lost revenue for both City Hotel and Resort Hotel – roughly 37% of all reservations in the dataset were cancelled. Price (ADR) is a strong driver of cancellation behaviour, City Hotel is more exposed to cancellations than Resort Hotel, January is the highest-risk month, and Online Travel Agent / Group bookings are the most cancellation-prone channels. By acting on pricing strategy, targeted discounts, seasonal promotions and channel management, both hotels can reduce their cancellation rates and improve revenue efficiency.